

Design Pattern 2: Factory Method Pattern

Exercise 2: Implementing the Factory Method Pattern

Scenario:

You are developing a document management system that needs to create different types of documents (e.g., Word, PDF, Excel). Use the Factory Method Pattern to achieve this.

Steps:

Create a New Java Project:

Create a new Java project named `FactoryMethodPatternExample`.

Define Document Classes:

Create interfaces or abstract classes for different document types such as `WordDocument`, `PdfDocument`, and `ExcelDocument`.

Create Concrete Document Classes:

Implement concrete classes for each document type that implements or extends the above interfaces or abstract classes.

Implement the Factory Method:

Create an abstract class `DocumentFactory` with a method `createDocument()`.

Create concrete factory classes for each document type that extends `DocumentFactory` and implements the `createDocument()` method.

Test the Factory Method Implementation:

Create a test class to demonstrate the creation of different document types using the factory method.